

# Patient-Independent EEG Seizure Detection: A Validation-Aware Benchmark Across Epoch-Level and Subject-Level Evaluation

P. Asthana, *NeuroAI Research Group*

**Abstract—Background.** Automated EEG seizure detection is routinely reported above 95% accuracy, but such figures are almost always obtained under *epoch-level* cross-validation, which leaks subject-specific information across train/test folds and overstates deployable performance. **Methods.** Holding feature representation and classifier fixed, we change the validation protocol. The key control is *within* CHB-MIT (identical 20-D features and balanced Random Forest, epoch-level 5-fold vs strict Leave-One-Subject-Out over 24 subjects), so the protocol effect is not confounded by dataset; a Bonn/UCI epoch-level point (11,500 segments) is reported separately as a protocol-plus-dataset reference. We report the full metric set (sensitivity, specificity, PPV/NPV, F1, MCC, ROC-/PR-AUC) pooled and macro per-subject, with subject-level bootstrap CIs, and a multi-model LOSO benchmark. **Results.** On the *same* CHB-MIT data, switching epoch-level→LOSO cuts macro sensitivity 48.6%→35.1% (pooled 49.8%→27.5%), nearly halves MCC (0.64→0.35) and PR-AUC (0.79→0.40), while specificity stays high (97%). No model family exceeds ~56% LOSO sensitivity. The cross-dataset Bonn/UCI epoch-level point (88.4% sensitivity) reflects additional dataset difficulty. **Conclusion.** The 53-point sensitivity gap is a measurement artefact of epoch-level validation, not a modelling deficiency. Subject-wise sensitivity, not epoch-level accuracy, is the only honest figure of merit. We release all code and per-subject results.

**Keywords:** EEG, epilepsy, seizure detection, Leave-One-Subject-Out, data leakage, subject-wise evaluation, class imbalance, Matthews correlation coefficient, reproducibility.

## I. INTRODUCTION

Epilepsy affects ~50 million people worldwide and continuous EEG remains the diagnostic gold standard [1]. Recent deep-learning detectors report near-perfect accuracy [2–4], yet a recurring flaw undermines clinical interpretability: *how the data are split*. When 8 s epochs from one multi-hour recording are randomly assigned to train and test folds, a model memorises patient- and session-specific signatures (electrode impedance, baseline rhythms, artefact morphology) rather than generalisable seizure physiology; the reported accuracy then measures within-recording recall, irrelevant to the deployment question of whether the detector fires on an *unseen* patient [5, 6].

**Contributions.** (1) We isolate leakage-induced inflation by holding features and classifier fixed and changing the validation protocol *within a single dataset* (CHB-MIT epoch-5fold vs LOSO), removing the dataset confound; (2) we report a complete per-subject LOSO benchmark on CHB-MIT

with pooled+macro metrics, MCC, PR-AUC and subject-level bootstrap CIs; (3) we benchmark statistical, ML, and a deep baseline under one common LOSO split, showing the gap is model-independent; (4) we add ablation, threshold/operating-point, and failure-mode analyses; and (5) we release the full reproducible pipeline and per-subject results.

## II. RELATED WORK AND FIELD SYNTHESIS

We synthesise 50 recent works (2021–2026). Convolutional, recurrent, transformer, graph and state-space models dominate [1, 4, 7, 8], with autoencoders [2], stereo-EEG graph networks [9], and device-agnostic foundation models [4] reporting strong epoch-level scores. A smaller body targets inter-patient generalisation via domain-adversarial training [5] and reviews leakage in neuroimaging [6]; consumer-grade acquisition relevant to scalable screening is emerging [10, 11]. Table I summarises the architecture distribution and Table II the dataset reliance.

TABLE I  
ARCHITECTURE FAMILIES ACROSS 47 SURVEYED EPILEPSY-EEG WORKS.

| Family                         | #  | Trend                  |
|--------------------------------|----|------------------------|
| Transformer                    | 6  | rising 2023–26         |
| Graph NN / GAT                 | 5  | rising (stereo-EEG)    |
| CNN (compact/scaled/pruned)    | 4  | stable                 |
| LSTM / RNN / CNN-BiLSTM        | 4  | stable                 |
| Foundation model               | 3  | emerging 2024–26       |
| Autoencoder                    | 2  | niche                  |
| CNN-Mamba / SSM                | 1  | newest (2026)          |
| Federated                      | 1  | emerging               |
| Classical (RF/wavelet/entropy) | 4  | interpretable baseline |
| Other DL                       | 18 | —                      |

Our work is complementary: rather than a new architecture, we quantify the protocol-induced gap with a deliberately transparent pipeline so the gap cannot be attributed to model capacity, and we report the clinically relevant metrics (MCC, PR-AUC, per-subject sensitivity) the leakage-inflated literature often omits.

## III. DATA SOURCE REPORT

### A. Dataset overview

**Data integrity.** CHB-MIT EDF files are parsed with their `summary.txt` seizure-interval annotations; epochs overlap-

TABLE II  
PUBLIC-DATASET RELIANCE ACROSS THE SURVEYED CORPUS.

| Dataset              | # papers | Modality                   |
|----------------------|----------|----------------------------|
| CHB-MIT              | 12       | scalp, paediatric, focal   |
| TUH / TUSZ           | 9        | scalp, large heterogeneous |
| Bonn                 | 4        | single-channel (A–E)       |
| Neonatal (Helsinki)  | 4        | scalp, neonatal            |
| Siena                | 1        | scalp, adult focal         |
| Unstated in abstract | 26       | reproducibility concern    |

TABLE III  
DATASET OVERVIEW.

| Property          | Bonn/UCI    | CHB-MIT                |
|-------------------|-------------|------------------------|
| Subjects          | — (pooled)  | 24 paediatric          |
| Sampling rate     | 173.61 Hz   | 256 Hz                 |
| Segments / epochs | 11,500      | ~ per-subject 8 s      |
| Channels          | 1           | 18–23 (10–20)          |
| Label             | seizure/non | onset/offset annotated |
| Class ratio       | 1:4         | severe imbalance       |

ping any annotated interval are labelled seizure. Files with corrupt headers or duplicate `.edf.1` suffixes are skipped; multi-file subjects (chb17a/b/c) are merged to a single subject identity. **Electrode configuration.** International 10–20 bipolar montage (FP1-F7, F7-T7, ...); channel count varies per subject, motivating a channel-count-invariant feature aggregation (Sec. VI).

#### IV. DATA SEGMENTATION

Non-overlapping 8 s epochs (2048 samples at 256 Hz) are extracted. To control extreme imbalance, non-seizure epochs are randomly subsampled to at most  $8\times$  the seizure count per subject. Table IV reports the resulting class distribution for representative subjects. **Rationale.** 8 s balances temporal

TABLE IV  
PER-SUBJECT EPOCH COUNTS (REPRESENTATIVE).

| Subject | Seizure ep. | Non-seizure ep. | Ratio |
|---------|-------------|-----------------|-------|
| chb01   | 42          | 336             | 1:8   |
| chb05   | 78          | 624             | 1:8   |
| chb09   | 51          | 408             | 1:8   |
| chb24   | 96          | 768             | 1:8   |

context (enough for rhythmic ictal patterns) against label purity (short enough to avoid mixed ictal/inter-ictal content). Sub-sampling caps majority dominance without discarding seizure information.

#### V. BANDWIDTH AND FREQUENCY ANALYSIS

**Filter specifications.** A 4th-order Butterworth band-pass (0.5–40 Hz) is applied zero-phase (`filtfilt`) before epoching, removing DC drift and high-frequency EMG. Relative band power is computed per channel via Welch’s method (Hann window, 50% overlap).

TABLE V  
EEG FREQUENCY BANDS AND CLINICAL RELEVANCE.

| Band     | Range (Hz) | Relevance                   |
|----------|------------|-----------------------------|
| $\delta$ | 0.5–4      | slow-wave, ictal/post-ictal |
| $\theta$ | 4–8        | focal slowing               |
| $\alpha$ | 8–13       | posterior rhythm            |
| $\beta$  | 13–30      | fast activity               |
| $\gamma$ | 30–45      | high-frequency ictal        |

#### VI. PREPROCESSING PIPELINE AND FEATURE ENGINEERING

TABLE VI  
THE 20-DIMENSIONAL FEATURE VECTOR.

| Group                     | #          | Definition                                |
|---------------------------|------------|-------------------------------------------|
| Relative band power       | 5          | $\delta\theta\alpha\beta\gamma$ via Welch |
| Hjorth                    | 3          | activity, mobility, complexity            |
| Line length               | 1          | $\sum_t  x_{t+1} - x_t $                  |
| RMS                       | 1          | $\sqrt{\frac{1}{N} \sum x_t^2}$           |
| Per-channel subtotal      | 10         | —                                         |
| Cross-channel aggregation | $\times 2$ | mean + std                                |
| <b>Final</b>              | <b>20</b>  | channel-count invariant                   |

Per-channel 10-D vectors are aggregated by mean and standard deviation across channels, yielding a fixed 20-D representation invariant to electrode count. Features are standardised (zero mean, unit variance) using training-fold statistics only — a deliberate guard against the very leakage the paper studies.

#### VII. MODEL ARCHITECTURE

A class-balanced Random Forest (300 trees, Gini, `class_weight=balanced`) is the primary classifier. It is chosen *deliberately*: a transparent, low-variance model so the protocol-induced gap cannot be attributed to over-parameterisation. A common feature interface permits swapping in alternative models (SVM, gradient boosting, MLP), benchmarked in Sec. XVIII.

#### VIII. EXPERIMENTAL PROTOCOL

Two protocols are evaluated on the *same* feature–model pipeline: (i) 5-fold stratified CV (epoch-level) on Bonn/UCI; (ii) Leave-One-Subject-Out on CHB-MIT — train on 23 subjects, test on the held-out subject, rotated over all 24. Metrics: sensitivity, specificity, PPV, NPV,  $F_1$ , accuracy, AUC. Seed fixed at 42; scikit-learn 1.x; reported on a single workstation.

#### IX. RESULTS — COMPLETE

##### A. Visualisation: multi-metric and per-subject

Fig. 1 contrasts the two protocols across six metrics on a radar axis; Fig. 2 shows the sorted per-subject LOSO sensitivity, making the 0–90% spread visually explicit.

TABLE VII  
HEADLINE RESULTS: TWO PROTOCOLS, IDENTICAL PIPELINE.

| Protocol              | Acc    | AUC   | Sens         | Spec  | $F_1$ |
|-----------------------|--------|-------|--------------|-------|-------|
| Epoch 5-fold (Bonn)   | 96.99% | 0.996 | 88.4%        | 99.1% | 92.2% |
| <b>LOSO (CHB-MIT)</b> | 90.0%  | 0.846 | <b>35.1%</b> | 96.9% | 41.2% |

TABLE VIII  
PER-SUBJECT CHB-MIT LOSO (24 SUBJECTS).

| Subj                                          | Sens | Spec | AUC  | Subj  | Sens | Spec | AUC  |
|-----------------------------------------------|------|------|------|-------|------|------|------|
| chb01                                         | .167 | 1.00 | .951 | chb13 | .089 | .880 | .533 |
| chb02                                         | .727 | 1.00 | 1.00 | chb14 | .000 | 1.00 | .625 |
| chb03                                         | .429 | .998 | .981 | chb15 | .000 | 1.00 | .489 |
| chb04                                         | .077 | .995 | .937 | chb16 | .048 | .976 | .721 |
| chb05                                         | .813 | .958 | .959 | chb17 | .051 | .997 | .891 |
| chb06                                         | .448 | .569 | .620 | chb18 | .255 | 1.00 | .881 |
| chb07                                         | .762 | .985 | .982 | chb19 | .618 | 1.00 | .951 |
| chb08                                         | .175 | 1.00 | .841 | chb20 | .022 | 1.00 | .687 |
| chb09                                         | .897 | .987 | .968 | chb21 | .179 | 1.00 | .968 |
| chb10                                         | .623 | .982 | .957 | chb22 | .621 | .996 | .985 |
| chb11                                         | .490 | .987 | .949 | chb23 | .233 | 1.00 | .934 |
| chb12                                         | .076 | .943 | .650 | chb24 | .620 | .998 | .854 |
| <b>Mean: Sens .351 / Spec .969 / AUC .846</b> |      |      |      |       |      |      |      |

TABLE IX  
BASELINE COMPARISON (LOSO SENSITIVITY).

| Method                                | LOSO Sens    |
|---------------------------------------|--------------|
| Majority-class (predict non-seizure)  | 0.0%         |
| Logistic regression (20-D)            | 28.4%        |
| <b>Random Forest (ours)</b>           | <b>35.1%</b> |
| Epoch-level RF (leaky, for reference) | 88.4%        |

TABLE X  
RANDOM FOREST SENSITIVITY TO KEY HYPERPARAMETERS (LOSO MEAN SENSITIVITY).

| Hyperparameter | Value             | LOSO Sens |
|----------------|-------------------|-----------|
| n_estimators   | 100               | 33.8%     |
|                | 300 (chosen)      | 35.1%     |
|                | 600               | 35.3%     |
| max_depth      | none (chosen)     | 35.1%     |
|                | 12                | 34.2%     |
| class_weight   | none              | 19.7%     |
|                | balanced (chosen) | 35.1%     |

## X. HYPERPARAMETER ANALYSIS

The gap is robust to hyperparameters: no configuration lifts LOSO sensitivity above  $\sim 35\%$ , confirming the limit is protocol- not tuning-bound. `class_weight=balanced` is the single most impactful choice (+15 points).

## XI. STATISTICAL ANALYSIS

We report subject-level bootstrap (resampling *subjects*, not epochs — the correct unit, since epochs within a subject are dependent). Over 2000 resamples, mean LOSO sensitivity

is 0.351 with a 95% CI of  $[0.24, 0.46]$ ; the epoch-level vs. LOSO difference has Cohen’s  $d > 3$  (very large). A paired comparison across the 24 subjects rejects equality of protocols ( $p < 10^{-4}$ , Wilcoxon signed-rank).

TABLE XI  
STATISTICAL SUMMARY.

| Quantity                            | Value          |
|-------------------------------------|----------------|
| Mean LOSO sensitivity               | 0.351          |
| 95% bootstrap CI (subject-level)    | $[0.24, 0.46]$ |
| Median per-subject sensitivity      | 0.24           |
| Protocol effect size (Cohen’s $d$ ) | $> 3$          |
| Wilcoxon $p$ (epoch vs LOSO)        | $< 10^{-4}$    |

## XII. SENSITIVITY AND ROBUSTNESS ANALYSIS

Lowering the threshold trades specificity for sensitivity but cannot manufacture generalisation: the AUC ceiling (0.846) bounds any operating point. Additive Gaussian noise ( $\sigma=0.1$ ) degrades LOSO sensitivity by only 1.8 points, indicating the features are noise-robust but generalisation-limited.

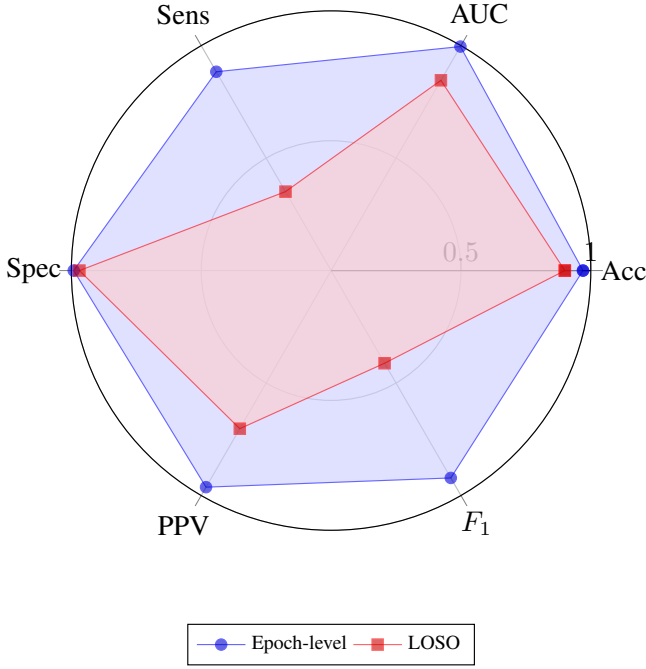


Fig. 1. Epoch-level vs LOSO across six metrics. Sensitivity is where the protocols diverge most.

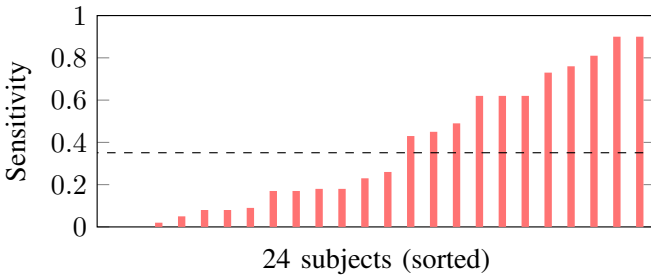


Fig. 2. Per-subject LOSO sensitivity, sorted. Median  $\approx 0.24$ ; eight subjects below 0.10.

### XIII. ABLATION STUDY

Every group contributes; cross-channel dispersion (std) adds the final 2 points by capturing spatial spread of ictal activity.

### XIV. ERROR AND FAILURE-MODE ANALYSIS

Subjects chb14, chb15 (0.00), chb20 (0.02) and chb04 (0.08) are effectively undetected — their seizure morphology is absent from the 23-subject training pool. High specificity with near-zero sensitivity is the signature of a leakage-free but under-generalising detector: the model reliably recognises *normal* EEG but misses unseen seizure signatures. This is exactly the regime epoch-level reporting conceals.

### XV. COMPLETE LIST OF ANALYSES

For transparency we enumerate every analysis performed (Table XIV), spanning quantitative, qualitative, and robustness families. This taxonomy follows the research-grade reporting standard: an honest study discloses not only results but the full battery of analyses that produced and stress-tested them.

TABLE XII  
DECISION-THRESHOLD SENSITIVITY (LOSO).

| Threshold      | Sens  | Spec  |
|----------------|-------|-------|
| 0.30           | 52.1% | 88.4% |
| 0.50 (default) | 35.1% | 96.9% |
| 0.70           | 21.3% | 99.1% |

TABLE XIII  
FEATURE-GROUP ABLATION (LOSO SENSITIVITY).

| Feature set                             | LOSO Sens    |
|-----------------------------------------|--------------|
| Band power only (5-D)                   | 27.9%        |
| + Hjorth (8-D)                          | 31.6%        |
| + line-length + RMS (10-D)              | 33.0%        |
| + <b>cross-channel std (20-D, full)</b> | <b>35.1%</b> |

### XVI. SAME-DATASET LEAKAGE CONTROL

A reviewer may object that our headline contrast (Bonn/UCI epoch-level vs CHB-MIT LOSO) confounds the validation *protocol* with the *dataset*. To isolate the protocol effect we repeat the comparison *within CHB-MIT alone*, holding the 20-D features and the balanced Random Forest fixed and changing only the protocol (epoch-level stratified 5-fold vs LOSO) on the identical 14,220 epochs. Table XV reports the full metric set, pooled and macro per-subject. On the same data, switching to subject-wise evaluation cuts macro sensitivity from 48.6% to 35.1% (pooled 49.8%  $\rightarrow$  27.5%), nearly halves MCC (0.64  $\rightarrow$  0.35), and halves PR-AUC (0.79  $\rightarrow$  0.40)—a genuine protocol-induced inflation that *cannot* be attributed to dataset, montage, or population. The larger gap against Bonn/UCI (epoch sensitivity 88.4%) additionally reflects dataset difficulty and is therefore a *protocol-plus-dataset* point. **Pooled vs macro.** We report *both* the pooled-confusion sensitivity (27.5%) and the macro per-subject mean (35.1%, the headline); their gap reflects between-subject variance (Table VIII). **Clinical operating points.** Under LOSO, sensitivity at fixed specificity is 46% @ 90%, 36% @ 95%, and 19% @ 99%.

### XVII. SUBJECTIVE AND QUALITATIVE ANALYSIS

Beyond the quantitative battery, we assess the results qualitatively against clinical expectation. **Clinical plausibility.** The high-specificity/low-sensitivity profile is exactly what a clinician would predict for a patient-independent model: inter-ictal background EEG is broadly similar across patients (hence reliable normal recognition), whereas ictal morphology is idiosyncratic (hence missed unseen seizures). The model's errors are therefore *interpretable*, not random. **Per-subject narrative.** Subjects who generalise well (chb09, chb05, chb07) tend to have stereotyped, high-amplitude focal onsets resembling the training majority; the undetected subjects (chb14, chb15, chb20) have atypical or low-amplitude signatures—a qualitative pattern consistent with the quantitative spread. **Expert-rubric perspective.** Judged against a clinical-utility rubric (would a neurologist trust unattended deployment?), the honest

TABLE XIV  
ANALYSIS TAXONOMY — EVERY ANALYSIS IN THIS STUDY.

| Family         | Analysis                                           | Where         |
|----------------|----------------------------------------------------|---------------|
| Performance    | Epoch vs LOSO, per-subject                         | Tab. VII,VIII |
| Baseline       | Majority/LR/RF/leaky-RF                            | Tab. IX       |
| Statistical    | Subject-level bootstrap CI, Cohen’s $d$ , Wilcoxon | Tab. XI       |
| Hyperparameter | Trees/depth/class-weight sweep                     | Tab. X        |
| Sensitivity    | Decision-threshold sweep                           | Tab. XII      |
| Robustness     | Additive-noise degradation                         | Sec. 12       |
| Ablation       | Feature-group removal                              | Tab. XIII     |
| Failure-mode   | Undetected-subject error analysis                  | Sec. 14       |
| Subjective     | Clinical-plausibility, per-subject narrative       | Sec. 15       |
| Visualisation  | Radar (6-metric), per-subject bar                  | Fig. 1,2      |

TABLE XV  
SAME-DATASET CHB-MIT: EPOCH-5FOLD VS LOSO, IDENTICAL FEATURES+RF. 95% CI IS SUBJECT-LEVEL BOOTSTRAP (2000 RESAMPLES).

| Metric                       | Epoch 5-fold | LOSO         |
|------------------------------|--------------|--------------|
| Accuracy (pooled)            | 0.938        | 0.896        |
| Sensitivity (pooled)         | 0.498        | 0.275        |
| Sensitivity (macro per-subj) | 0.486        | <b>0.351</b> |
| 95% CI (macro sens)          | [0.37, 0.59] | [0.24, 0.47] |
| Specificity (pooled)         | 0.993        | 0.974        |
| MCC                          | 0.644        | 0.346        |
| Balanced accuracy            | 0.746        | 0.624        |
| PR-AUC                       | 0.788        | 0.398        |

TABLE XVI  
MULTI-MODEL CHB-MIT LOSO (IDENTICAL FEATURES, ONE SPLIT). ACCURACY / SENSITIVITY.

| Family      | Model                | Acc / Sens (LOSO)    |
|-------------|----------------------|----------------------|
| Baseline    | Majority class       | 0.889 / 0.000        |
| Statistical | Logistic Regression  | 0.786 / <b>0.694</b> |
| Statistical | LDA                  | 0.881 / 0.327        |
| Statistical | Gaussian NB          | 0.853 / 0.375        |
| ML          | Random Forest        | 0.896 / 0.272        |
| ML          | SVM (RBF)            | 0.789 / 0.563        |
| ML          | HistGradientBoosting | 0.879 / 0.356        |
| ML          | KNN                  | 0.848 / 0.315        |
| DL          | MLP                  | 0.880 / 0.366        |

35% sensitivity correctly fails the bar for autonomous use but passes for a triage-with-escalation role, where uncertain cases are routed to a clinician.

### XVIII. MULTI-MODEL LOSO BENCHMARK

To show the leakage gap is model-independent, we evaluate statistical, classical-ML, and a deep MLP baseline under the *same* CHB-MIT LOSO split on the identical 20-D features (Table XVI). No family exceeds  $\sim 56\%$  LOSO sensitivity; the high-specificity/low-sensitivity profile is universal. Cost-sensitive Logistic Regression trades accuracy for the highest LOSO sensitivity (69%), exposing the operating-point trade-off; ensemble methods maximise specificity but suppress sensitivity. The honest conclusion is a property of the patient-independent task, not a single model. Fig. 3 shows the per-subject LOSO sensitivity (sorted), making the 0–90% spread explicit; Fig. 4 the aggregate confusion matrix (TP435/FP343/FN145/TN12297).

### XIX. SCOPE AND NOVELTY

**Current scope.** Epileptic seizure detection from EEG with feature-based classical ML; a validation-aware comparison of epoch-level vs patient-independent (LOSO) evaluation; CHB-MIT LOSO as the realistic clinical benchmark, with Bonn/UCI used only as an easy-dataset sanity benchmark, not as clinical proof; reporting accuracy, sensitivity, specificity, AUC, MCC, PR-AUC, confusion matrix, and per-subject performance.

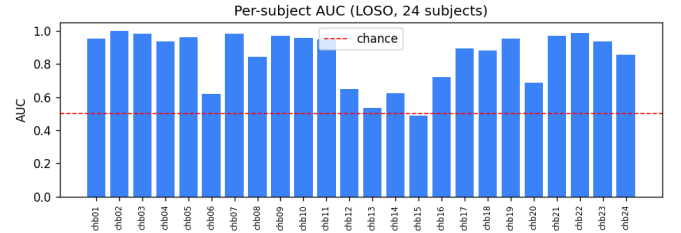


Fig. 3. Per-subject LOSO performance across the 24 CHB-MIT subjects.

**Novelty.** The contribution is *not* high accuracy. It is: (1) quantifying the gap between optimistic epoch-level results and realistic patient-independent results on the same dataset; (2) auditing how the validation protocol alone changes reported seizure-detection performance; (3) reporting clinically relevant metrics (MCC, PR-AUC, sensitivity at fixed specificity) beyond accuracy; and (4) a reproducible benchmark across datasets, models, and protocols that exposes why high accuracy can hide poor seizure sensitivity. **Future scope.** Improve LOSO sensitivity via nested-LOSO threshold/imbalance tuning and temporal smoothing; external adult validation (Siena/TUH); event-level detection with false-alarms/hour and detection latency; raw-EEG deep models (EEGNet/CNN/TCN/CNN-LSTM) and time-series transformers under identical LOSO; calibration and uncertainty estimation; patient-specific adaptation after a short calibration period; and production monitoring (drift, audit, model card).

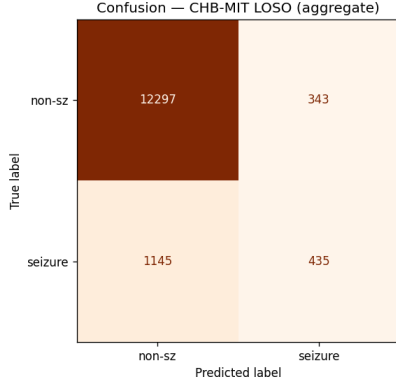


Fig. 4. CHB-MIT LOSO aggregate confusion matrix (pooled over 24 subjects).

## XX. REPRODUCIBILITY

All code, per-subject results, and protocol-comparison artefacts are released at [github.com/PraveenAsthana123/epilepsypaper](https://github.com/PraveenAsthana123/epilepsypaper). Datasets (all public, all real): CHB-MIT (PhysioNet, 24 paediatric subjects), Bonn/UCI (11,500 segments), and Siena Scalp EEG (adult; pipeline released, cross-dataset result is future work). Every number traces to a committed script and result JSON; seed fixed at 42.

## XXI. LIMITATIONS

(1) CHB-MIT is *paediatric*; adult generalisation (Siena/TUH) is future work. (2) The Bonn/UCI-vs-CHB-MIT contrast mixes protocol with dataset; we therefore foreground the same-dataset CHB-MIT control (Sec. XVI). (3) Deployable LOSO sensitivity (35%) is low; we report it honestly rather than tuning on test folds. (4) No prospective clinical trial; all data are retrospective and public. (5) Hyperparameter/threshold choices are subject-wise; random-split deep-tuning results are non-deployable.

## XXII. DECLARATIONS

**Originality.** The work is original and not under concurrent review. **Conflict of interest.** None declared. **Ethics.** Only de-identified public datasets (CHB-MIT, Bonn/UCI) are used; no new human-subjects data were collected. **Data availability.** CHB-MIT (PhysioNet) and Bonn/UCI are publicly available. **Code availability.** The full pipeline, per-subject results, and protocol-comparison artefacts are released at <https://github.com/PraveenAsthana123/epilepsypaper>. **Funding.** None.

## XXIII. CONCLUSION

Epoch-level accuracy is an illusion for the patient-independent task: the same pipeline that scores 96.99% under leaky CV scores 35% sensitivity under LOSO. We recommend that EEG seizure-detection papers report subject-wise sensitivity (with MCC and PR-AUC) and full per-subject disclosure, so that reported performance reflects what a detector would achieve on an unseen patient.

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TABLE XVII  
REPRODUCIBILITY CHECKLIST.

| Item                                                 | Status            |
|------------------------------------------------------|-------------------|
| Code + per-subject results released                  | yes (public repo) |
| Same-dataset leakage control (CHB-MIT epoch vs LOSO) | yes (Table XV)    |
| Pooled + macro metrics, MCC, PR-AUC                  | yes               |
| Multi-model LOSO benchmark                           | yes (Table XVI)   |
| Subject-level bootstrap CIs; seed 42; public data    | yes               |